

Replication of Wu 2018

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Question: Which words are most predictive of a poster's gender status on Economics Job Market Rumors (EJMR)?

Data: Scaped EJMR posts

Method: LASSO logistic regression to identify predictive words

- Let $\mathbf{w}_i$ denote a vector of word counts for post i (excluding gender classifiers)
- Estimate LASSO-regularized logistic model for probability that post is about a woman

Model:

$$\hat{\theta}_\lambda = \arg \min_{\theta} -\log \left(\prod_{i=1}^N P(\text{Female}_i | \mathbf{w}_i) \right) + \lambda \|\theta\|_1$$

where $\|\theta\|_1 = \sum_{j \geq 1} |\theta^j|$

Replication Discussion

- Wu 2018 provides compelling evidence that online economics discussions differ when discussing Woman versus Men
- Most predictive words were those referring to personal lives, family, sex/gender issues
- Male predictive words involved professional and academic career status

Replication Difficulties:

- Outdated packages and computational efficiency
 - .py files used outdated numpy functions
 - Large sample size (several million) leading to lengthy runtime
 - May need to parallelize processes on multiple cores
- Overall adjusting the.py files required modifying 3 lines each giving me access to all required files to build tables from.

Table 1 Wu (2018)

word	ME	word	ME
hotter	0.4223134	homo	-0.3030683
pregnant	0.3227177	testosterone	-0.1953370
plow	0.2772726	chapters	-0.1893582
marry	0.2753558	satisfaction	-0.1867459
hot	0.2710024	fiickers	-0.1808531
marrying	0.2600944	macroeconomics	-0.1800874
pregnancy	0.2541431	cuny	-0.1798895
attractive	0.2450691	thrust	-0.1687640
beautiful	0.2398092	nk	-0.1650142
breast	0.2265265	macro	-0.1631897

- Robustness check with modified subsample next

Table 2 Wu (2018)

word	ME_pronoun	word	ME_pronoun
pregnancy	0.2922947	knocking	-0.3294384
hotter	0.2893404	testosterone	-0.2043009
pregnant	0.2583185	blog	-0.1825698
hp	0.2380338	hateukbro	-0.1764642
vagina	0.2280140	adviser	-0.1745582
breast	0.2201753	hero	-0.1738850
plow	0.2190030	cuny	-0.1732126
shopping	0.2073728	handsome	-0.1661303
marry	0.2072572	mod	-0.1660732
gorgeous	0.2008608	homo	-0.1599676

Extension: Random Forest Model

Motivation: Complement Wu (2018) LASSO approach with a non-linear model

Method:

- 100 decision trees, each trained on random sample of posts (replacement)
- At each node: selects best word from ~ 98 words (of 9,540 total) using Gini impurity
- Final prediction is just majority vote across all trees

Key Findings:

- Similar predictive words as LASSO $\rightarrow$ gendered language is primarily linear/additive
- LASSO's interpretability (fixed coefficients) outweighs Random Forest's complexity
- Random Forest lacks clear effect sizes; predictions emerge from different decision paths

References

-  Breiman, L. (2001). *Random forests*. Machine Learning, 45(1), 5-32.
-  Wu, A. (2018). *Gendered language on the economics job market rumors forum*. AEA Papers and Proceedings, 108, 175-179.